TONU JAMES

APPLIED AI | DATA SCIENCE | SOFTWARE ENGINEERING

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LinkedIn | GitHub | Live AI Portfolio

PROFESSIONAL PROFILE

MSc graduate in Human-Centred Artificial Intelligence from Technological University Dublin, with a strong foundation in Machine Learning, Computer Vision, and Data Analytics. Experienced software engineer with a professional background in enterprise application development, complemented by recent academic AI research and hands-on development of deployed Generative AI and Edge AI applications. Seeking to transition into Applied AI and Data Science, combining technical knowledge, practical AI implementation skills, and established software engineering experience.

EDUCATION

Master of Science in Human-Centred Artificial Intelligence 2025

Technological University Dublin, Ireland | Second Class Honours

Master of Technology in Computer Science (Image Processing) 2014

Cochin University of Science and Technology (CUSAT), India | First Class with Distinction

Bachelor of Technology in Computer Science and Engineering 2012

College of Engineering Kidangoor (CUSAT), India | First Class

TECHNICAL SKILLS

Python & Data Science: Python, Pandas, NumPy, scikit-learn, Matplotlib, Seaborn; data preprocessing, exploratory data analysis, visualisation, time-series forecasting (SARIMA, Holt-Winters, Prophet).

Machine Learning & Computer Vision: CNNs, YOLO/YOLOX, Siamese Networks, KNN, VGG19 transfer learning, Keras, OpenCV, model evaluation, Explainable AI (Grad-CAM).

Generative AI & NLP: RAG, document chunking, embeddings, semantic search, vector retrieval; ChromaDB, FAISS, FastEmbed, Hugging Face embeddings, Groq LLM APIs; LangChain (intermediate).

Software Development & APIs: Git/GitHub, FastAPI, REST APIs, React, JavaScript, SAP ABAP, Java, OData.

AI Deployment & Automation: Docker, GitHub Actions, self-hosted runners, GHCR, ONNX Runtime, GStreamer, PySide6, Raspberry Pi 5.

SELECTED APPLIED AI PROJECTS

Full-Stack AI Portfolio with Integrated RAG Chatbot 2026

Live portfolio | GitHub repository

- Designed and deployed a full-stack AI portfolio integrating a React-based frontend with a Python FastAPI backend, enabling visitors to interact with an AI-powered chatbot through natural-language queries.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using structured Markdown documents, text chunking, FastEmbed embeddings, ChromaDB vector similarity search and Groq-hosted LLM inference to generate context-aware responses grounded in portfolio information.
- Deployed the React frontend on Vercel and Python backend on Render, integrating REST API communication to deliver a publicly accessible, end-to-end Generative AI application.

Reusable Edge AI Model & Application Deployment | Raspberry Pi 5 2026

Application repository | CD pipeline

- Designed and implemented a reusable Edge AI continuous delivery framework using GitHub Actions and a Raspberry Pi 5 self-hosted runner, enabling independent deployment of ONNX models and containerised GUI/Web applications.
- Developed separate Model CD and Application CD workflows, incorporating model versioning, SHA-256 integrity verification, deployment manifests, rollback support and Docker-based application delivery through GitHub Container Registry (GHCR).
- Demonstrated the framework by developing and deploying a Dockerised YOLO object-detection desktop application integrating GStreamer camera pipelines, a PySide6 GUI and ONNX Runtime for live detection on Raspberry Pi 5.

PROFESSIONAL EXPERIENCE

Technological University Dublin | Software Development Lab Assistant (Part-time) Mar–May 2024

- Supported Python object-oriented programming laboratories for approximately 50 second-year undergraduate students, explaining concepts, debugging student code and guiding practical exercises.

SAP Labs India | Software Developer Dec 2015–Aug 2023

- Led a four-member technical team supporting the SAP Multi-Bank Connectivity Connector for 100+ customer organisations; coordinated incident investigation, debugging, fixes and stakeholder communication.
- Developed and maintained customer-facing enterprise CRM and product-lifecycle applications, including banking workflows for RBC and Standard Bank of South Africa and packaging-related workflows for Tetra Pak.
- Built an ABAP/OData data-transformation utility to extract and reshape transactional and master data for migration to a non-SAP system.
- Explored YOLO object detection, few-shot product matching and autoencoder-based document-image processing through internal AI proof-of-concept activities (not production deployments).

Tata Consultancy Services | Associate Developer Dec 2014–Nov 2015

- Developed Java-based interactive financial risk reporting functionality for a Credit Suisse investment-banking application working with live-streaming information.

AI RESEARCH & ACADEMIC PROJECTS

Regional and Demographic Bias in Pedestrian Detection | MSc Thesis 2025

- Created and curated a 2,173-image Indian road-scene dataset with 5,416 final pedestrian bounding boxes, including manual annotation and quality filtering of source images.
- Evaluated a pretrained YOLOX detector across Indian and European street conditions using IoU, miss-rate metrics and statistical analysis; identified miss rates of 28.27% versus 6.56%, respectively.
- Investigated demographic subgroup differences and discussed dataset representation, fairness, model reliability and responsible AI evaluation.

Face Similarity & Explainable AI | ACM Conference Poster 2024

- Built and evaluated Siamese networks for face similarity, improving reported accuracy from 49.61% with a custom CNN to 95.4% using VGG19 transfer learning; compared KNN classification accuracy (91.25%).
- Investigated model explainability using Grad-CAM for the Siamese network and pixel-perturbation saliency maps for KNN.

  Publication: Exploring Trade-offs Between Black-Box and Glass-Box Models in Face Similarity: Siamese Networks vs. KNN. ACM HCAIep 2024. DOI 10.1145/3701268.3701284

Dublin Pedestrian Footfall Forecasting | Time-Series Study 2023

- Integrated 2021–2023 Dublin City Council sensor records, standardised locations, handled missing data and resampled hourly counts into a 1,035-day forecasting series.
- Compared Holt-Winters, SARIMA and Prophet, using RMSE to evaluate forecasts; SARIMA recorded the lowest reported RMSE (24,163) on the studied dataset.